

In a 4-bit computer system, you can find the two's complement of numbers like 5 and -2 by following these steps:

For 5 (in 4-bit binary):

- Decimal 5 in 4-bit binary is 0101.
- To find the two's complement, first invert all the bits:
- Inverting 0101 gives 1010.
- Add 1 to the inverted binary representation:
- Adding 1 to 1010 gives 1011.

So, the two's complement of 5 in a 4-bit system is 1011, which is equivalent to -5 in decimal.

For -2 (in 4-bit binary):

- Decimal -2 in 4-bit binary is typically represented as a signed 4-bit number. To find its two's complement, we follow the steps:
- The binary representation of -2 is 1110 (assuming 4 bits).
- Inverting the bits gives 0001.
- Adding 1 to 0001 gives 0010.

So, the two's complement of -2 in a 4-bit system is 0010, which is equivalent to 2 in decimal.



how to calculate -2 in binary

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To represent -2 in binary using two's complement notation, you typically follow these steps:

- Start with the positive binary representation of the absolute value of the number. In this case, the absolute value of -2 is 2, which in binary is 10.
- Invert (flip) all the bits in the binary representation obtained in step 1. Change all 0s to 1s and all 1s to 0s.
- Add 1 to the inverted binary representation.

Let's calculate -2 in binary using two's complement notation:

- Start with the binary representation of 2, which is 10.
- Invert the bits:
- Flipping 10 gives 01.
- Add 1 to the inverted representation:
- Adding 1 to 01 gives 10.

So, -2 in binary using two's complement notation is 10.



do the above in 4 bit system

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Certainly! To represent -2 in binary using two's complement notation in a 4-bit system, follow these steps:

- Start with the positive binary representation of the absolute value of the number. In this case, the absolute value of -2 is 2, which in binary is 0010 in a 4-bit system.
- Invert (flip) all the bits in the binary representation obtained in step 1. Change all 0s to 1s and all 1s to 0s.
- Add 1 to the inverted binary representation.

Let's calculate -2 in binary using a 4-bit two's complement notation:

- Start with the binary representation of 2, which is 0010 in a 4-bit system.
- Invert the bits:
- Flipping 0010 gives 1101.
- Add 1 to the inverted representation:
- Adding 1 to 1101 gives 1110.

So, -2 in binary using two's complement notation in a 4-bit system is 1110.